

15. Count Characters, Words and Lines in a File

```
#include <stdio.h>

#include <ctype.h>

int main()
{
    FILE *fp;
    char filename[50];
    char ch;
    int characters = 0;
    int words = 0;
    int lines = 0;
    int inword = 0;
    printf("Enter file name: ");
    scanf("%s", filename);

    fp = fopen(filename, "r");

    if (fp == NULL)
    {
        printf("File cannot be opened.\n");
        return 0;
    }

    while ((ch = fgetc(fp)) != EOF)
    {
        characters++;

        if (ch == '\n')
```

```

        lines++;

    if (isspace(ch))

        inword = 0;

    else if (inword == 0)

    {

        words++;

        inword = 1;

    }

}

fclose(fp);

printf("\nCharacters = %d\n", characters);

printf("Words    = %d\n", words);

printf("Lines    = %d\n", lines);

return 0;

}

```

OUTPUT

```

Lexical Analyzer
-----
Number of Characters = 53
Number of Words      = 8
Number of Lines      = 3
-----
Process exited after 1.029 seconds with return value 0
Press any key to continue . . . |

```